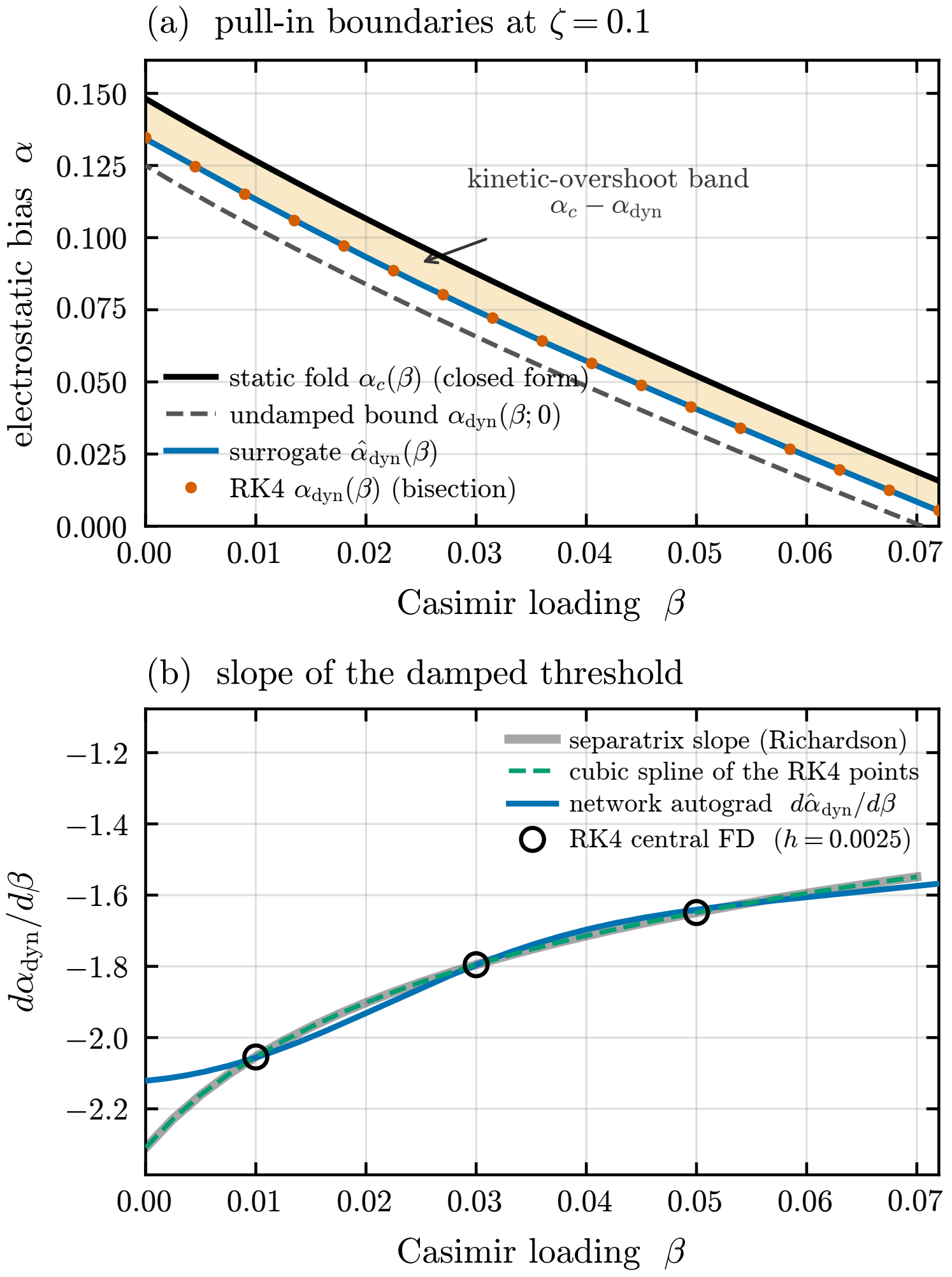
(a) pull-in boundaries at $\zeta = 0.1$
electrostatic bias α
0.150
0.125
0.100
0.075
0.050
0.025
0.000
kinetic-overshoot band
$\alpha_c - \alpha_{\mathrm{dyn}}$
static fold $\alpha_c(\beta)$ (closed form)
undamped bound $\alpha_{\mathrm{dyn}}(\beta; 0)$
surrogate $\hat{\alpha}_{\mathrm{dyn}}(\beta)$
RK4 $\alpha_{\mathrm{dyn}}(\beta)$ (bisection)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Casimir loading β
(b) slope of the damped threshold
$d\alpha_{\mathrm{dyn}}/d\beta$
−1.2
−1.4
−1.6
−1.8
−2.0
−2.2
separatrix slope (Richardson)
cubic spline of the RK4 points
network autograd $d\hat{\alpha}_{\mathrm{dyn}}/d\beta$
RK4 central FD ($h = 0.0025$)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
Casimir loading β